

5.0 Introduction [Numerical Integration and Differentiation]

Section 21.1, 21.2, 22.2, 22.4, 23.1, 23.2

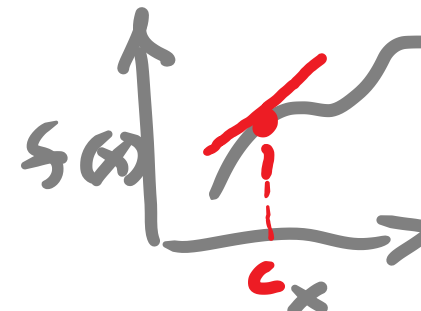
There are many cases where we need to evaluate the derivative or integral (of a function or a data set) numerically.

Particularly important for discretized data (i.e. measured or experimental data), or complex functions.

$$\int_a^b f(x) dx = \text{AREA UNDER CURVE}$$



$$\frac{d f(x)}{dx} = \text{SLOPE OF THE CURVE}$$



* for measured or tabulated data, we are limited by the number of points (measurement frequency)

5.0 Introduction [Numerical Integration and Differentiation]

Numerical Integration Methods

- Trapezoid rule
 - Can be applied to discrete data (evenly spaced or unevenly spaced data)
 - Can be applied to a continuous function
 - Easy to implement, but needs a large number of intervals for good accuracy
- Simpson's 1/3 rule
 - Can be applied to discrete data (requires evenly spaced data, and even number of intervals)
 - Can be applied to a continuous function
 - Works well for smooth functions or data, improved accuracy over Trapezoid rule
- Romberg Integration (integration of equations)
 - Only for a continuous function
 - Function must be continuously differentiable, points are equally-spaced
 - Extrapolation of integral from two estimates yields improved accuracy with reduced calculations compared to Trapezoid and Simpson's
- Gauss Quadrature (integration of equations)
 - Only for a continuous function
 - Achieves higher order accuracy with smaller number of points and smaller number of calculations compared to other methods

5.0 Numerical Integration [General]

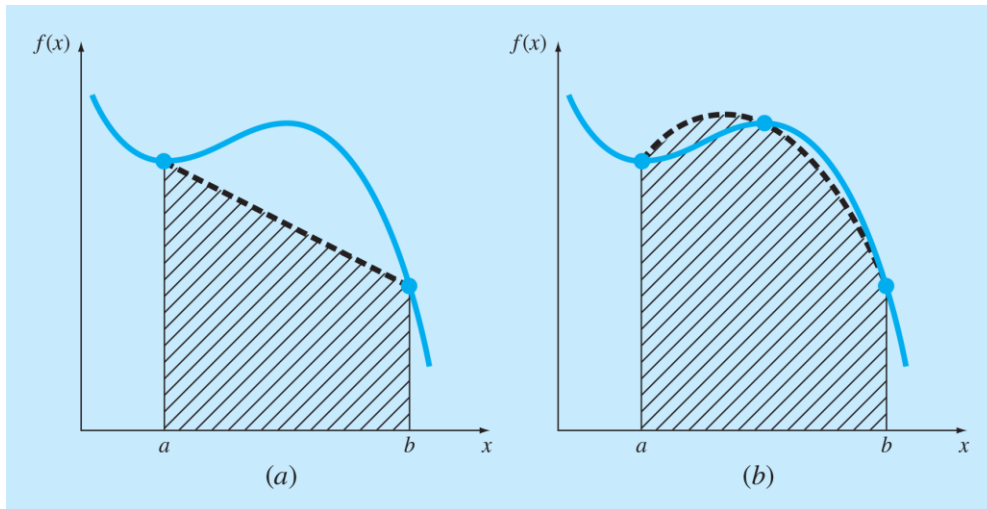
$$I = \int_a^b f(x) dx \cong \int_a^b f_n(x) dx$$

Consider a function $f(x)$ defined over a range $a \leq x \leq b$

$$f_n(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + a_nx^n$$

1. Divide the domain $a \leq x \leq b$ into evenly-spaced intervals
2. Estimate the function or tabulated data using a curve (e.g. linear, polynomial)
3. Estimate the area under the curve as the summation of the area of the individual intervals.

A polynomial describing the function over the interval



Simple linear representation of the curve for each interval

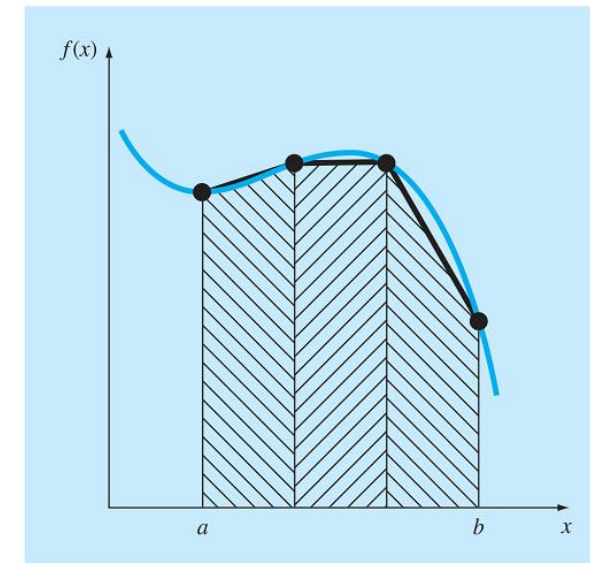


FIGURE 21.2 The approximation of an integral by the area under three straight-line segments.

5.1 Numerical Integration [Trapezoid Rule – applied to one interval]

Section 21.1

Trapezoid rule – a subset of the Newton-Cotes closed (data at endpoints) formulations using a first order (linear) representation of the curve.

* replacing data or a complex function with a simple function

Consider a small interval between a and b

$$I = \int_a^b f(x) dx \cong \int_a^b f_1(x) dx \quad f_1(x) = f(a) + \frac{f(b) - f(a)}{b - a} (x - a)$$

$$I = \int_a^b \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right] dx$$

$$I = (b - a) \frac{f(a) + f(b)}{2}$$

Box 21.1 in Chapra shows the details

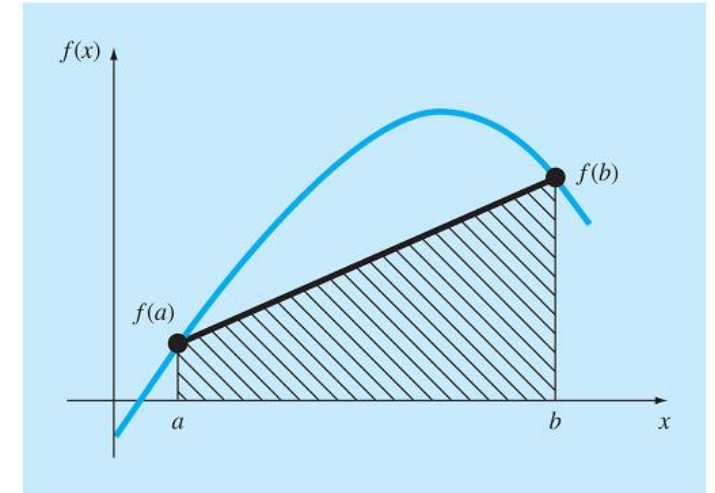
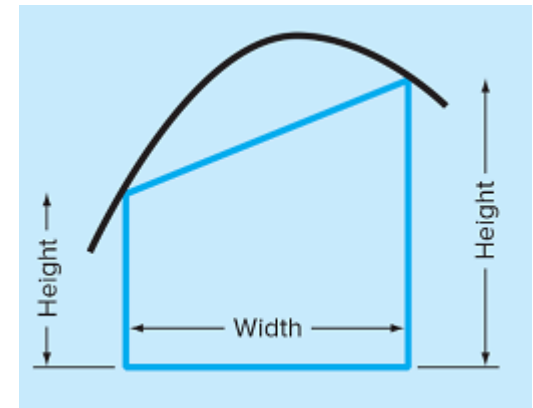


FIGURE 21.4 Graphical depiction of the trapezoidal rule.

$$I \cong (b - a) \times \text{average height}$$



5.1 Numerical Integration [Composite Trapezoid Rule]

The trapezoid rule can be applied to achieve improved accuracy by dividing into n intervals.

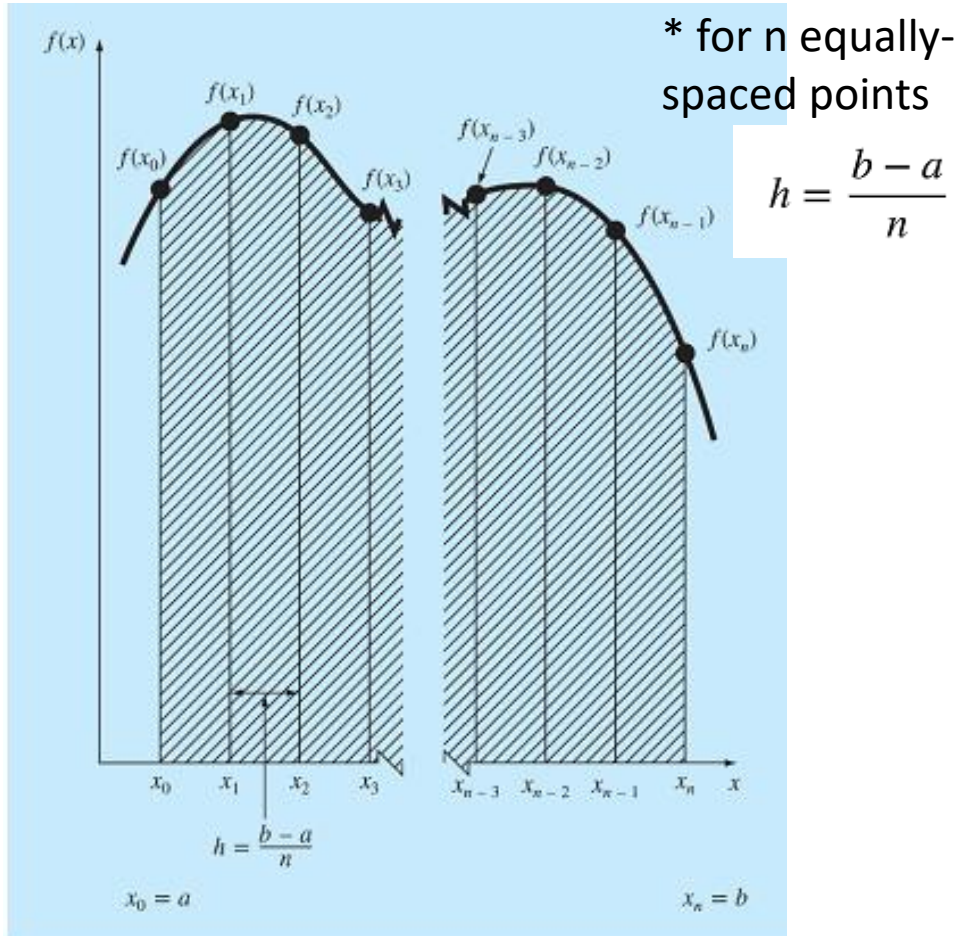


FIGURE 21.8 The general format and nomenclature for multiple-application integrals.

Total area (I) is the summation of each individual area.

$$I = \int_{x_0}^{x_1} f(x)dx + \int_{x_1}^{x_2} f(x)dx + \dots + \int_{x_{n-1}}^{x_n} f(x)dx$$

Using the trapezoid rule:

$$I = h \frac{f(x_0) + f(x_1)}{2} + h \frac{f(x_1) + f(x_2)}{2} + \dots + h \frac{f(x_{n-1}) + f(x_n)}{2}$$

And collecting similar terms:

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

← EASIER TO PROGRAM

or

$$I = \underbrace{(b-a)}_{\text{width}} \underbrace{\frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$

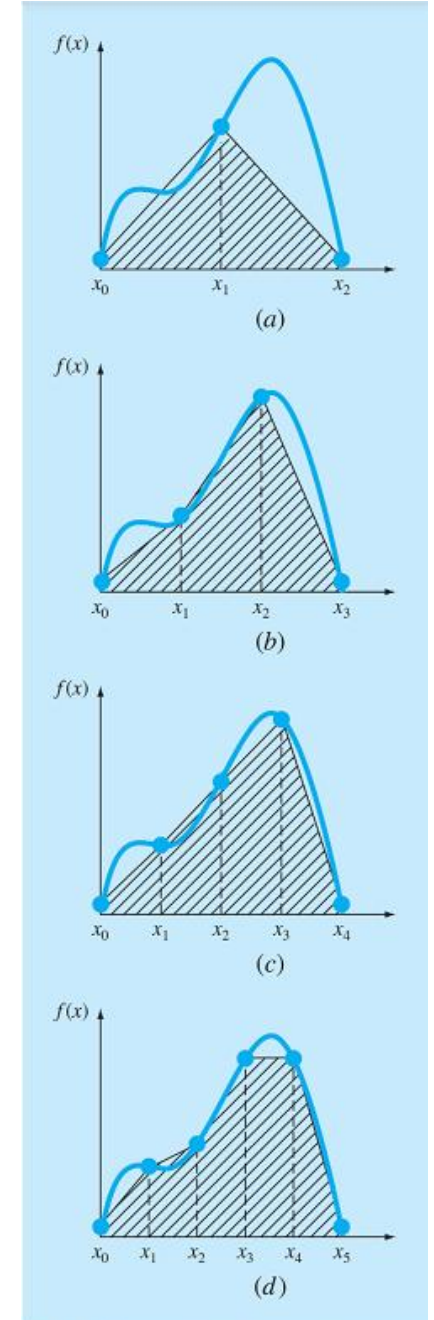
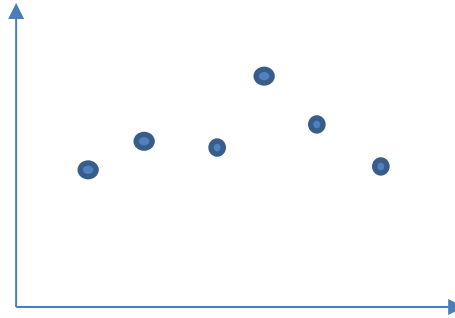


FIGURE 21.7 Illustration of the multiple-application trapezoidal rule. (a) Two segments, (b) three segments, (c) four segments, and (d) five segments.

5.1 Numerical Integration [Trapezoid Rule]

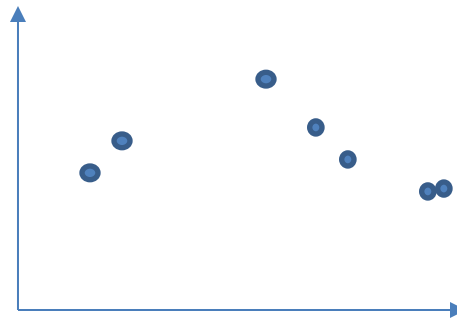
The trapezoid rule can be applied to evenly-spaced data:

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$



and unevenly-spaced (non-uniform grid) data:

$$I = h_1 \frac{f(x_0) + f(x_1)}{2} + h_2 \frac{f(x_1) + f(x_2)}{2} + \dots + h_n \frac{f(x_{n-1}) + f(x_n)}{2}$$



The trapezoid rule is one of the most often-used methods because:

- a) Easy to implement
- b) Can be used for both data and functions
- c) Data can be unevenly-spaced

We are just applying the trapezoid rule to each interval h_i

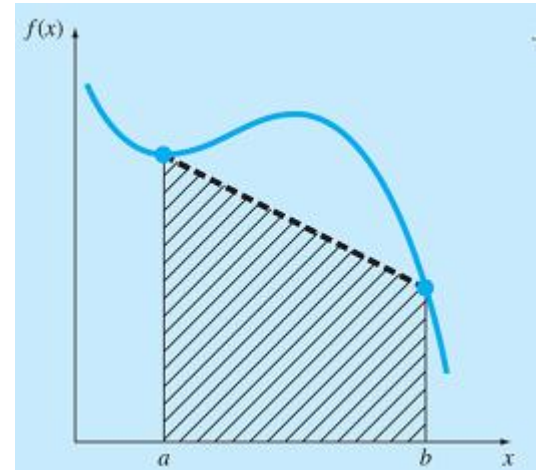
h_i = the width of segment i

Chapra recommends a hybrid approach – trapezoid for unequal spaced data, but use Simpson's 1/3 where data is equally spaced.

5.1 Numerical Integration [Trapezoid Rule - Accuracy]

In general, accuracy improves as we increase the number of intervals.

The error in applying the trapezoid rule can be determined by summing up the errors of each individual interval. Where ξ (pronounced zaai) lies somewhere in the interval from a to b .



The error for an individual interval is:
(the second derivative describes the curvature – for a linear function, $f''=0$ and the representation is exact).

$$E_t = -\frac{1}{12}f''(\xi)(b-a)^3$$

See box 21.2 for derivation

✓ or $\xi_i - \xi_{i+1}$

For n intervals

$$E_t = -\frac{(b-a)^3}{12n^3} \sum_{i=1}^n f''(\xi_i)$$

where $f''(\xi_i)$ is the second derivative at a point ξ_i located in segment i

$$\sum f''(\xi_i) \cong n\bar{f}''$$

$\bar{f}''$ is the mean or average value of the second derivative for the entire interval.

$$E_a = -\frac{(b-a)^3}{12n^2} \bar{f}''$$

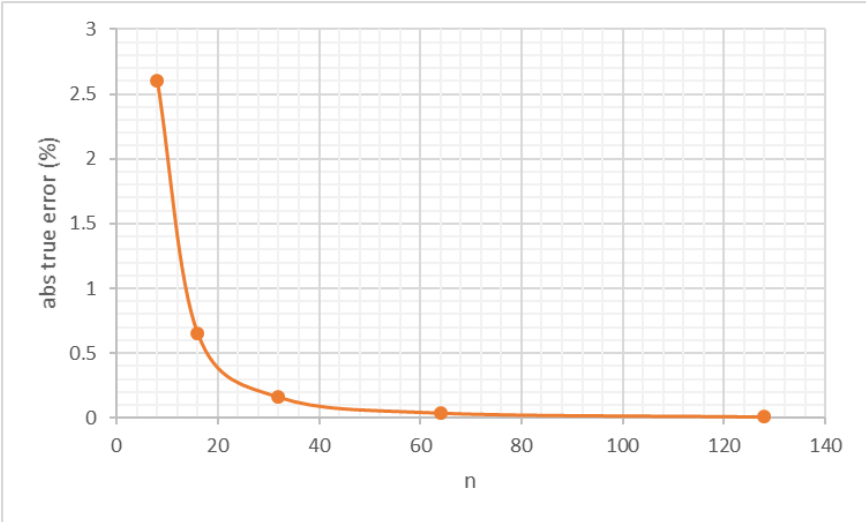
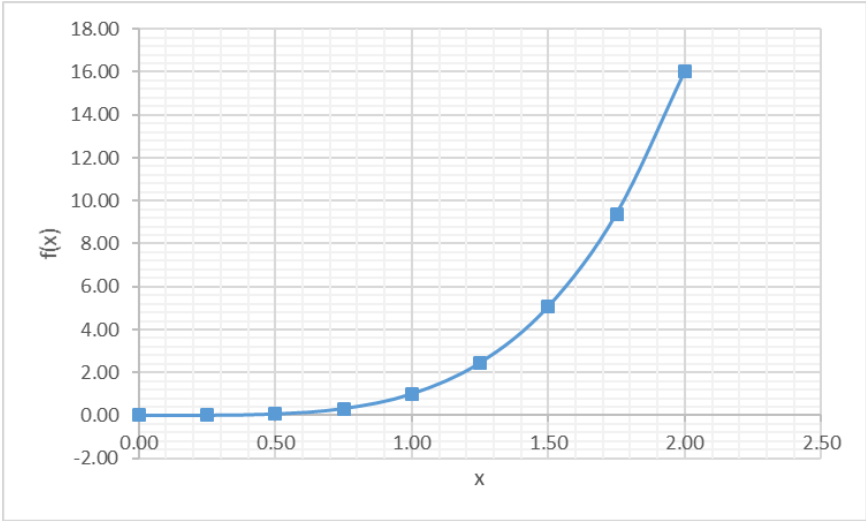
The approximate error (due to approx. of f'') is a function of $1/n^2$
So doubling the number of points, reduces the error by a factor of 4.

5.1 Numerical Integration [Trapezoid Rule]

* The curve is concave up, so the trapezoid method overestimates the area.

Example 5.1: Use the trapezoid rule to find the integral with n=8, 16, 32, 64 and 128 of $f(x)=x^4$ over the interval from $0 \leq x \leq 2$.

h=(b-a)/n	0.25		h=(b-a)/n	0.125		h=(b-a)/n	0.0625		h=(b-a)/n	0.03125		h=(b-a)/n	0.015625	
Trapezoid	6.566406		Trapezoid	6.44165		Trapezoid	6.410416		Trapezoid	6.402604		Trapezoid	6.400651	
et	2.600			0.651			0.163			0.041			0.010	
n=	8		n=	16		n=	32		n=	64		n=	128	
i	x	f(x)	i	x	f(x)	i	x	f(x)	i	x	f(x)	i	x	f(x)
0	0.00	0.00	0	0.00	0.00	0	0.00	0.00	0	0.00	0.00	0	0.00	0.00
1	0.25	0.00	1	0.13	0.00	1	0.06	0.00	1	0.03	0.00	1	0.02	0.00
2	0.50	0.06	2	0.25	0.00	2	0.13	0.00	2	0.06	0.00	2	0.03	0.00
3	0.75	0.32	3	0.38	0.02	3	0.19	0.00	3	0.09	0.00	3	0.05	0.00
4	1.00	1.00	4	0.50	0.06	4	0.25	0.00	4	0.13	0.00	4	0.06	0.00
5	1.25	2.44	5	0.63	0.15	5	0.31	0.01	5	0.16	0.00	5	0.08	0.00
6	1.50	5.06	6	0.75	0.32	6	0.38	0.02	6	0.19	0.00	6	0.09	0.00
7	1.75	9.38	7	0.88	0.59	7	0.44	0.04	7	0.22	0.00	7	0.11	0.00
8	2.00	16.00	8	1.00	1.00	8	0.50	0.06	8	0.25	0.00	8	0.13	0.00
			9	1.13	1.60	9	0.56	0.10	9	0.28	0.01	9	0.14	0.00
			10	1.25	2.44	10	0.63	0.15	10	0.31	0.01	10	0.16	0.00
			11	1.38	3.57	11	0.69	0.22	11	0.34	0.01	11	0.17	0.00
			12	1.50	5.06	12	0.75	0.32	12	0.38	0.02	12	0.19	0.00
			13	1.63	6.97	13	0.81	0.44	13	0.41	0.03	13	0.20	0.00
			14	1.75	9.38	14	0.88	0.59	14	0.44	0.04	14	0.22	0.00
			15	1.88	12.36	15	0.94	0.77	15	0.47	0.05	15	0.23	0.00
			16	2.00	16.00	16	1.00	1.00	16	0.50	0.06	16	0.25	0.00
						17	1.06	1.27	17	0.53	0.08	17	0.27	0.00



$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

$$\int_0^2 x^4 dx = \frac{1}{5} x^5 \Big|_0^2 = 6.4$$

Aside... Error Calculation using the trapezoidal rule

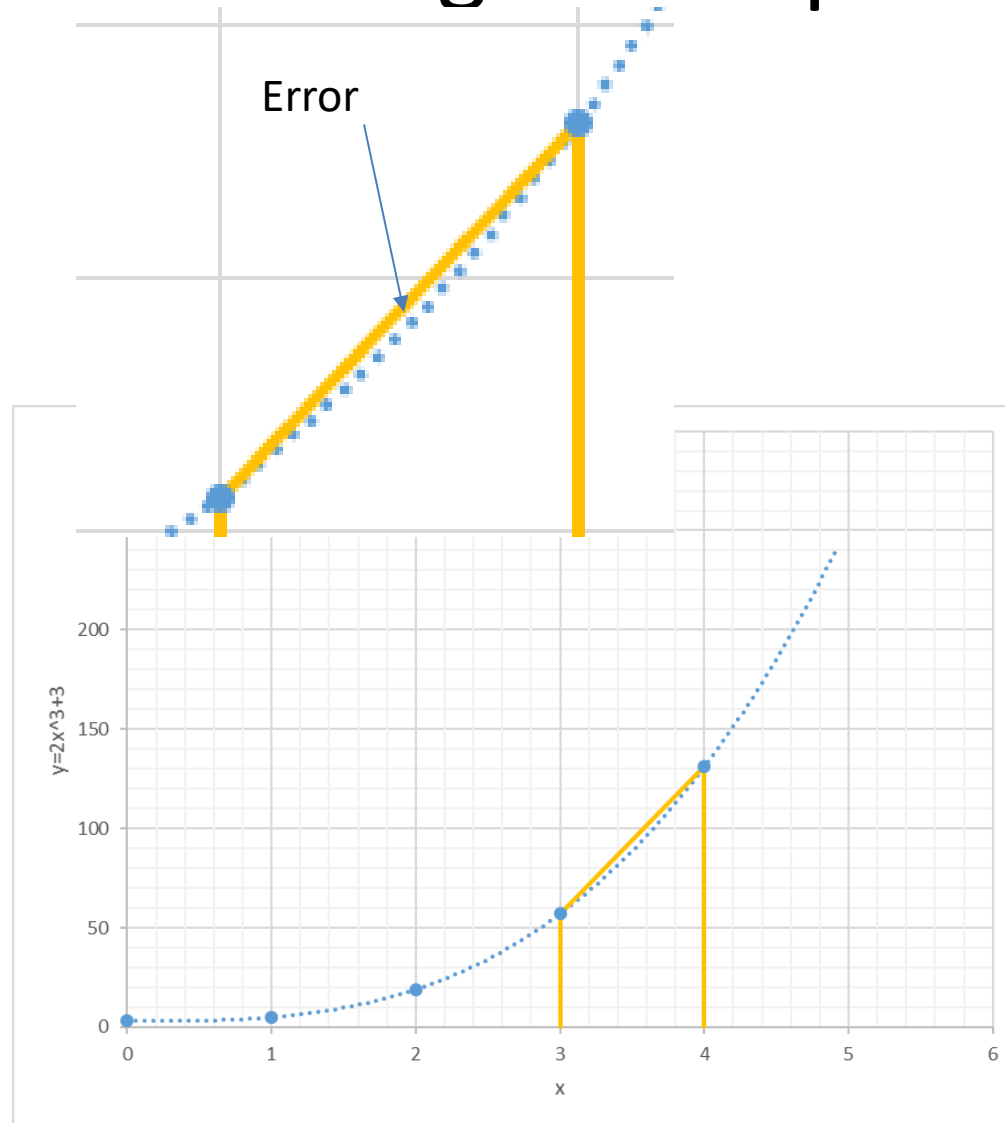
Function: $y = 2x^3 + 3$

$$y' = 6x^2$$

$$y'' = 12x$$

$$I = \frac{1}{2}x^4 + 3x$$

Exact integral from x=3 to x=4		
I=	90.5	
Trapezoid function from x=3 to 4		
3	0	Area (integral estimate)
3	57	94
4	131	Error = -3.5
4	0	(actual - estimate)



Approximate error

$$E_a = -\frac{(b-a)^3}{12n^2} \bar{f}''$$

b	4
a	3
n	1

$\bar{f}''$

$f'' = 12x$

$f''(3) =$	36
$f''(4) =$	48
$\bar{f}''$	42

$$E_a = -3.5$$

error is a function of $1/n^2$
alternatively a function of h^2

$$h = \frac{b-a}{n}$$

Aside...

2nd order Lagrange Interpolating Polynomial

- A reformulation of Newton's divided difference, that doesn't require calculating the divided difference

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad \Pi \text{ designates the "product of."}$$

- For a first order (linear n=1) Lagrange Polynomial

$$f_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

- For a 2nd order (n=2) Lagrange Polynomial

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

Recall for 2nd order Newton divided difference (section 4.4)

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

Linear Newton divided difference

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$$

5.2 Numerical Integration [Simpson's Rule]

Section 21.2

Simpson's rule – using higher order polynomials to improve the approximation of the function in each interval.

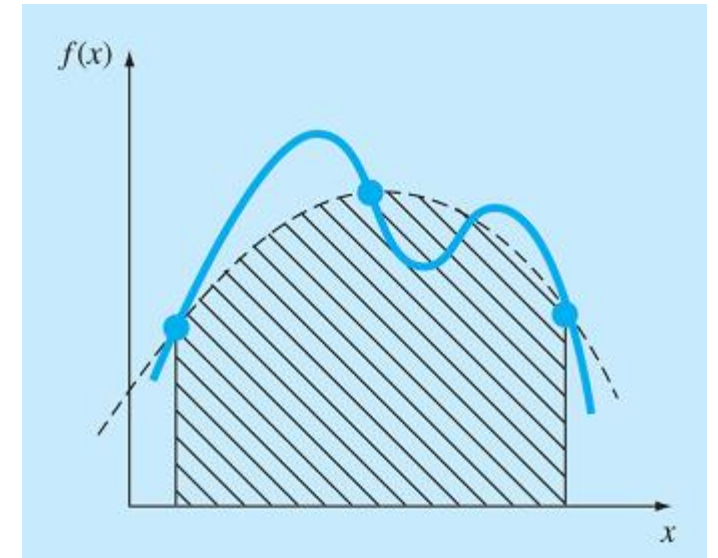
Simpson's 1/3 rule – connecting points using a quadratic function.

$$I = \int_a^b f(x) dx \cong \int_a^b f_2(x) dx$$

2nd ORDER

Using a 2nd order Lagrange polynomial interpolation, and after algebraic manipulation (section 21.2), the area for each interval is:

$$I \cong \underbrace{(b-a)}_{\text{width}} \underbrace{\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}}_{\text{Average height}}$$



This is the second Newton Cotes closed integration formulation

1/3 rule name -> h is divided by 3

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4 \sum_{i=1, 3, 5}^{n-1} f(x_i) + 2 \sum_{j=2, 4, 6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$

5.2 Numerical Integration [Composite Simpson's 1/3 Rule]

Section 21.2

For multiple intervals (* requires an even number of intervals or segments):

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4 \sum_{i=1, 3, 5}^{n-1} f(x_i) + 2 \sum_{j=2, 4, 6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$

Handwritten notes:
 ODD TERMS (pointing to the first sum)
 EVEN TERMS (pointing to the second sum)

The error is a function of $1/n^4$:

$$E_a = -\frac{(b-a)^5}{180n^4} \bar{f}^{(4)}$$

$\bar{f}^{(4)}$ is the average 4th derivative

So doubling the number of intervals 'n', reduces the error by a factor of 16 or (n^4)

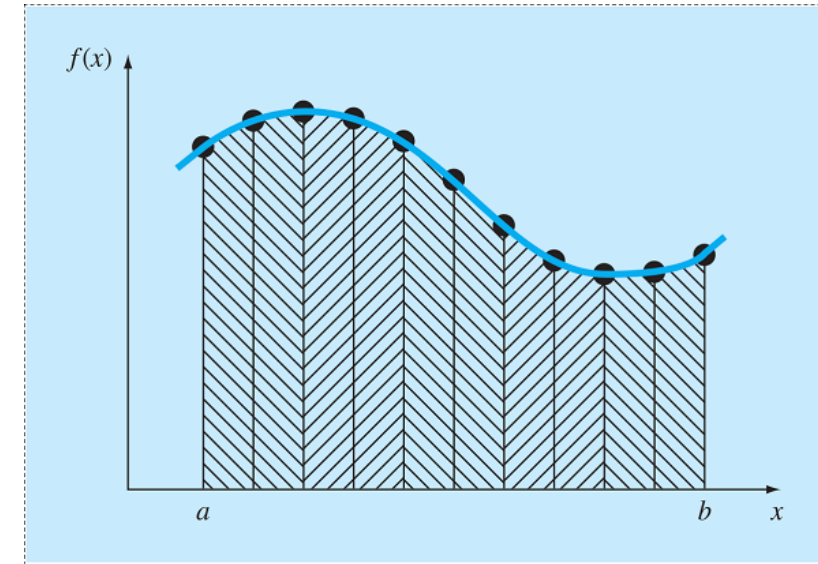


FIGURE 21.11 Graphical representation of the multiple application of Simpson's 1/3 rule. Note that the method can be employed only if the number of segments is even.

5.2 Numerical Integration [Simpson's 1/3 Rule]

Example 5.2: Use the trapezoid rule and Simpson's rule to find the integral with $n=8$ of $f(x)=x^4$ over the interval from $0 \leq x \leq 2$.

n=		8		
i	x	f(x)		
0	0.00	0.00		
1	0.25	0.00		
2	0.50	0.06		
3	0.75	0.32		
4	1.00	1.00		
5	1.25	2.44		
6	1.50	5.06		
7	1.75	9.38		
8	2.00	16.00		

h=(b-a)/n	0.25	Absolute Error	% true error
Trapezoid	6.566406	-0.16641	2.600
Simpsons 1/3	6.401042	-0.00104	0.016
Exact solution	6.4		
1/5 x^5			

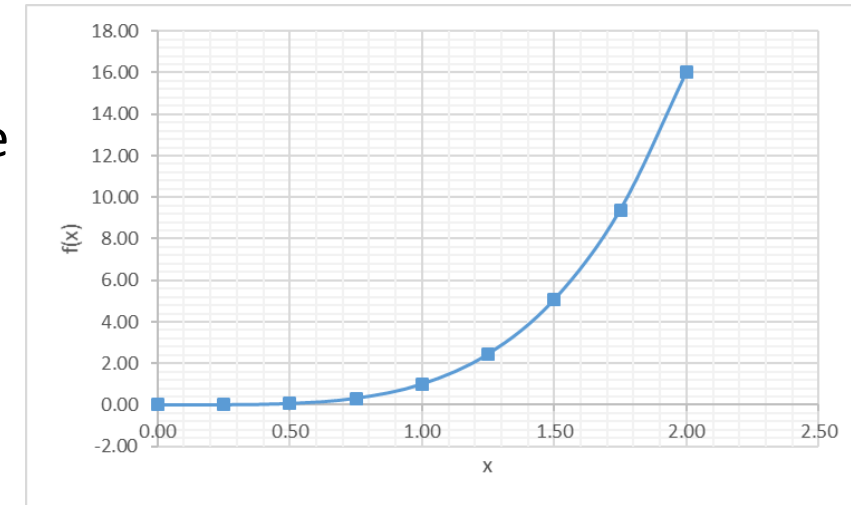
$$I_{\text{EXACT}} = I_{\text{APPROX}} + E$$

$$\epsilon_t = \text{abs} \left(\frac{\text{TRUE} - \text{APPROX}}{\text{TRUE}} \right) \times 100\%$$

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

$$I \cong \underbrace{\frac{(b-a)}{\text{Width}}}_{\text{Average height}} \underbrace{\left[f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n) \right]}_{3n}$$

$$I_{\text{exact}} = I_{\text{Simpsons}} + E$$



$$\int_a^b x^4 dx = \frac{1}{5} x^5 \Big|_0^2 = 6.4$$

* The curve is concave up, so the trapezoid method overestimates the area.

5.3 Numerical Integration [Romberg Integration of Functions]

Challenges with previous methods:

- Trapezoid and Simpson's rules can require a large number of terms.
- Increasing the number of terms can lead to round-off error.
- Simpson's rule requires an even number of intervals.

Romberg Integration – using two successive trapezoid rule estimates of the integral to obtain a 3rd more accurate value (using Richardson extrapolation).

Often achieved by halving the interval (doubling n).

Requires that the function be sufficiently differentiable over the evaluation interval.

Recall the integral evaluation using the trapezoid rule.

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

Double the number of intervals:

n is doubled

h is halved

5.3 Numerical Integration [Romberg – Richardson Extrapolation]

The error associated with the integral is:

$$I = I(h) + E(h) \quad h = \frac{b-a}{n} \quad * \text{ Trapezoid error estimates are of } O(h^2) \text{ (recall the } 1/n^2 \text{ term)}$$

Consider two estimates using different step sizes h_1 and h_2 , then:

$$I(h_1) + E(h_1) = I(h_2) + E(h_2)$$

Recall that the error associated with the trapezoidal rule could be approximated as:

$$E \cong -\frac{b-a}{12} h^2 \bar{f}'' \quad [\text{with } n = (b-a)/h]$$

Assuming that $\bar{f}''$ is constant (not dependent on step size), then the

ratio: $\frac{E(h_1)}{E(h_2)} \cong \frac{h_1^2}{h_2^2}$ or $E(h_1) \cong E(h_2) \left(\frac{h_1}{h_2} \right)^2$

Gives: $I(h_1) + E(h_2) \left(\frac{h_1}{h_2} \right)^2 \cong I(h_2) + E(h_2) \quad \longrightarrow \quad E(h_2) \cong \frac{I(h_1) - I(h_2)}{1 - (h_1/h_2)^2}$

* we have an estimate for the error in terms of the two integral estimations and their step sizes

5.3 Numerical Integration [Romberg – Richardson Extrapolation]

From:

$$I = I(h) + E(h) \qquad E(h_2) \cong \frac{I(h_1) - I(h_2)}{1 - (h_1/h_2)^2} \qquad I(h_1) + E(h_1) = I(h_2) + E(h_2)$$

Giving:

$$I \cong I(h_2) + \frac{1}{(h_1/h_2)^2 - 1} [I(h_2) - I(h_1)]$$

* It can be shown that the error of this estimate is $O(h^4)$, even though the estimate came from two trapezoid estimates with errors of $O(h^2)$.

Rearranging for $h_2=h_1/2$:

$$I \cong I(h_2) + \frac{1}{2^2 - 1} [I(h_2) - I(h_1)]$$

or

$$I \cong \frac{4}{3}I(h_2) - \frac{1}{3}I(h_1) \quad \Rightarrow \quad O(h^4)$$

* This is an improved estimate of the integral, extrapolated from two previous estimates.

5.3 Numerical Integration [Romberg Integration]

We can make progressive improvements as follows

$$I \cong \frac{4}{3}I(h_2) - \frac{1}{3}I(h_1) \Rightarrow O(h^4)$$

Similarly, combining two improved $O(h^4)$ integrals gives a result of $O(h^6)$

$$I \cong \frac{16}{15}I_m - \frac{1}{15}I_l \Rightarrow O(h^6)$$

I_m and I_l are the more and less accurate estimates

Combining two improved $O(h^6)$ integrals gives a result of $O(h^8)$

$$I \cong \frac{64}{63}I_m - \frac{1}{63}I_l \Rightarrow O(h^8)$$

In general:

$$I_{j,k} \cong \frac{4^{k-1}I_{j+1,k-1} - I_{j,k-1}}{4^{k-1} - 1}$$

where $I_{j+1,k-1}$ and $I_{j,k-1}$ are the more and less accurate integrals, respectively, and $I_{j,k}$ is the improved integral. The index k signifies the level of the integration, where $k = 1$ corresponds to the original trapezoidal rule estimates, $k = 2$ corresponds to $O(h^4)$, $k = 3$ to $O(h^6)$, and so forth. The index j is used to distinguish between the more $(j + 1)$ and the less (j) accurate estimates.

* I_m – more accurate – is the one with more terms/higher n value

$$|\varepsilon_a| = \left| \frac{I_{1,k} - I_{2,k-1}}{I_{1,k}} \right| 100\%$$

* Relative approximate error can be used as a stopping criterion

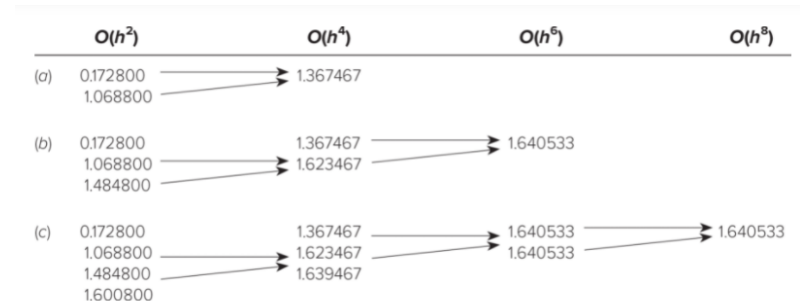


FIGURE 22.3 Graphical depiction of the sequence of integral estimates generated using Romberg integration. (a) First iteration. (b) Second iteration. (c) Third iteration.

5.3 Numerical Integration [Romberg Integration]

Example 5.3: (data from example 5.1 – integral of $f(x)=x^4$ over the interval from $0 \leq x \leq 2$)

Examlle 5.3 Romberg integration							
(info from example 5.1)							
n	l(h)						
	O(h ²)		O(h ⁴)		O(h ⁶)		O(h ⁸)
8	6.56640625		6.400065104		6.400000000000000000000000		6.400000000000000000000000
16	6.441650391		6.400004069		6.400000000000000000000000		6.400000000000000000000000
32	6.410415649		6.400000254		6.400000000000000000000000		6.400000000000000000000000
64	6.402604103		6.400000016				
128	6.400651038						

5.4 Numerical Integration of Functions [Gauss Quadrature]

Section 22.4

Recall, for the trapezoid rule, the base points are fixed and the error is the difference in area between the actual curve and the straight line. All methods considered to this point (trapezoid, Simpson's, Romberg) use the data measured points (equally spaced).

Gauss Quadrature [aka Gauss-Legendre]

Employs values positioned between the evaluation/measured points to achieve an improved estimate of the integral.

If we can 'balance' the positive and negative errors between the straight line approximation and the curve, we can achieve a better estimate of the integral.

Provides exact values of integrals for polynomials up to degree $2n - 1$ (for n points). e.g. for 2 data points, accurate for polynomial up to order 3

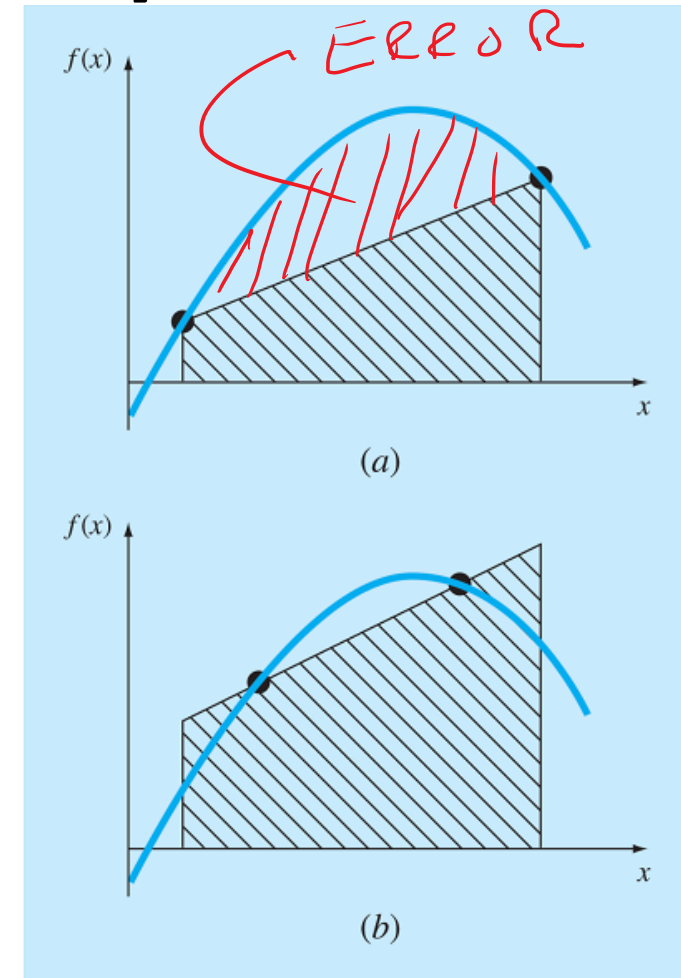


FIGURE 22.6 (a) Graphical depiction of the trapezoidal rule as the area under the straight line joining fixed end points. (b) An improved integral estimate obtained by taking the area under the straight line passing through two intermediate points. By positioning these points wisely, the positive and negative errors are balanced, and an improved integral estimate results.

5.4 Numerical Integration [Gauss Quadrature]

Two-point Gauss-Legendre Formula

Using the method of undetermined coefficients – the area (integral) can be approximated as:

$$I \cong c_0 f(x_0) + c_1 f(x_1)$$

Where c_0 and c_1 are unknown coefficients, and x_0 and x_1 are not fixed at the endpoints (4 unknowns requires 4 equations).

To develop 4 equations, we assume the 4 unknowns fit the integrals of:

- i) Constant function (horizontal line) ($y=\text{constant}$)
- ii) Linear function ($y=x$)
- iii) Parabolic function ($y=x^2$)
- iv) Cubic function ($y=x^3$)

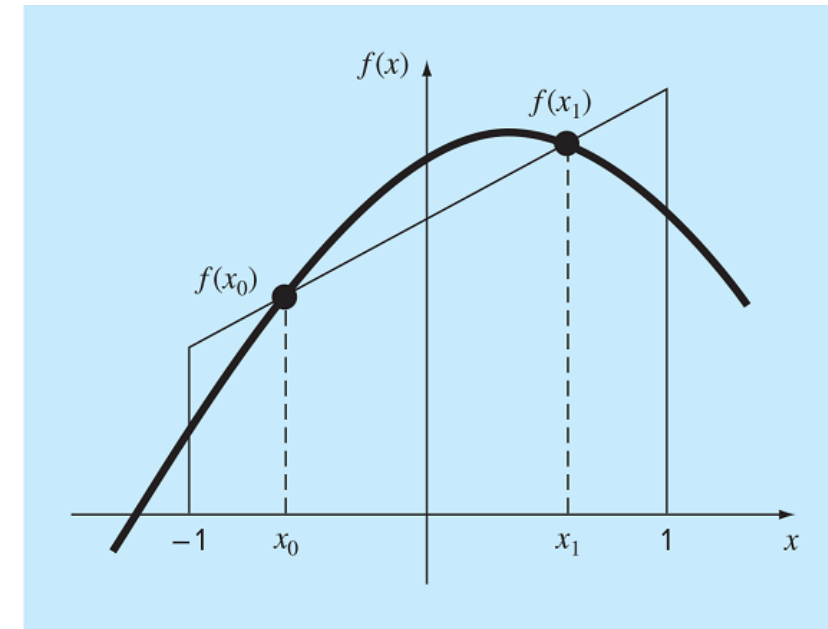


FIGURE 22.8 Graphical depiction of the unknown variables x_0 and x_1 for integration by Gauss quadrature.

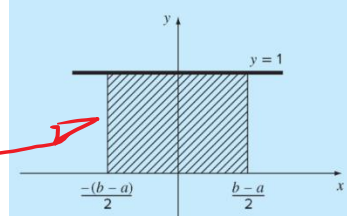
* Applying the equations i) and ii) we can reproduce the trapezoid rule, points are located at the ends of the interval

5.4 Numerical Integration [Gauss Quadrature]

Two-point Gauss-Legendre Formula (in normalized x_d space)

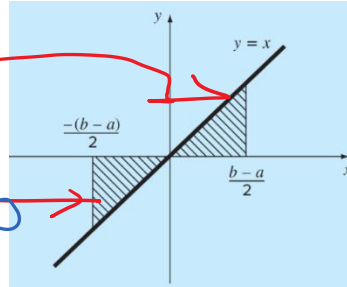
i) Constant function (horizontal line) ($y=\text{constant}$)

$$c_0 f(x_0) + c_1 f(x_1) = \int_{-1}^1 1 dx = 2$$



ii) Linear function ($y=x$)

$$c_0 f(x_0) + c_1 f(x_1) = \int_{-1}^1 x dx = 0$$

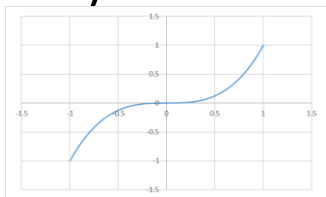


iii) Parabolic function ($y=x^2$)

$$c_0 f(x_0) + c_1 f(x_1) = \int_{-1}^1 x^2 dx = \frac{2}{3}$$

iv) Cubic function ($y=x^3$)

$$c_0 f(x_0) + c_1 f(x_1) = \int_{-1}^1 x^3 dx = 0$$



* Note that we are working in a space from $-1 \leq x_d \leq 1$

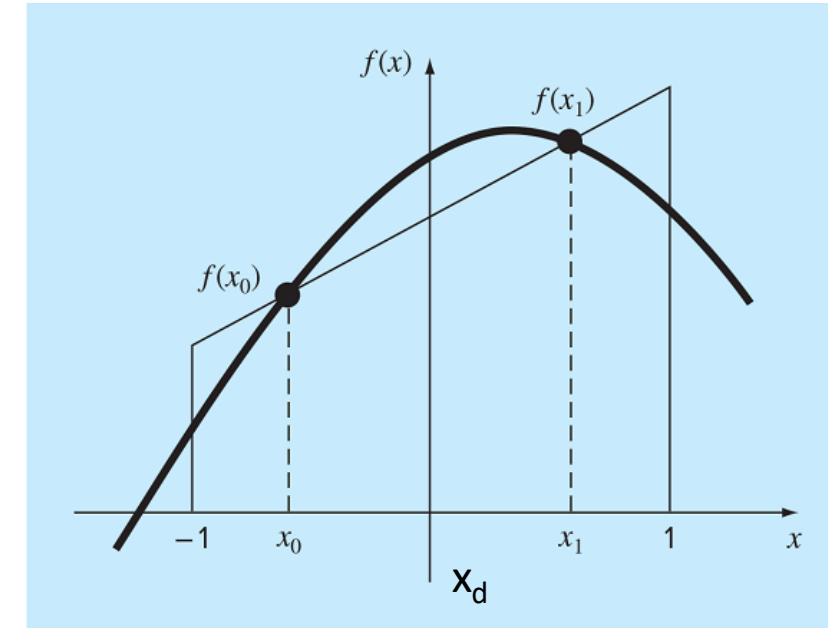


FIGURE 22.8 Graphical depiction of the unknown variables x_0 and x_1 for integration by Gauss quadrature.

$$\Rightarrow c_0 = c_1 = 1$$

$$x_0 = -\frac{1}{\sqrt{3}}$$

$$x_1 = +\frac{1}{\sqrt{3}}$$

* Integral = 0 for odd functions – areas sum to zero

5.4 Numerical Integration [Gauss Quadrature]

Two-point Gauss-Legendre Formula

$$c_0 = c_1 = 1$$

$$x_0 = -\frac{1}{\sqrt{3}} = -0.5773503 \dots$$

$$x_1 = \frac{1}{\sqrt{3}} = 0.5773503 \dots$$

$$I \cong f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

* For integration limits from -1 to +1
If the limits are different, the slope and intercept need to change.

* Moving the points to $\pm 1/\sqrt{3}$ gives a result that is third-order accurate.

Substitution, assume the general variable 'x' is a linear function of variable x_d (from -1 to 1).

$$x = a_0 + a_1 x_d$$

$x = a$ corresponds to $x_d = -1$

$x = b$ corresponds to $x_d = +1$

$$a = a_0 + a_1 (-1)$$

$$b = a_0 + a_1 (1)$$

Solving



$$a_0 = \frac{b + a}{2}$$

$$a_1 = \frac{b - a}{2}$$



$$x = \frac{(b + a) + (b - a)x_d}{2}$$

$$dx = \frac{b - a}{2} dx_d$$

* This mapping transforms the integration interval without affecting the value of the integral.

5.4 Numerical Integration [Gauss Quadrature]

Two-point Gauss-Legendre Formula example: integrate $f(x)=x^4$ over the interval from $0 \leq x \leq 2$

$$I = \int_0^2 x^4 dx$$

$$= \int_{-1}^1 (1+x_d)^4 dx_d$$

$$x = \frac{b+a}{2} + \frac{b-a}{2} x_d$$

$$= \frac{2+0}{2} + \frac{2-0}{2} x_d$$
$$= 1 + x_d$$

$$dx = \frac{b-a}{2} dx_d$$

$$= \frac{2-0}{2} dx_d$$
$$= dx_d$$

Apply Two Point
Gauss Quadrature

$$I = f(x_d = -\frac{1}{\sqrt{3}}) \\ + f(x_d = +\frac{1}{\sqrt{3}})$$

5.4 Numerical Integration [Gauss Quadrature]

Two-point Gauss-Legendre Formula example: integrate $f(x)=x^4$ over the interval from $0 \leq x \leq 2$

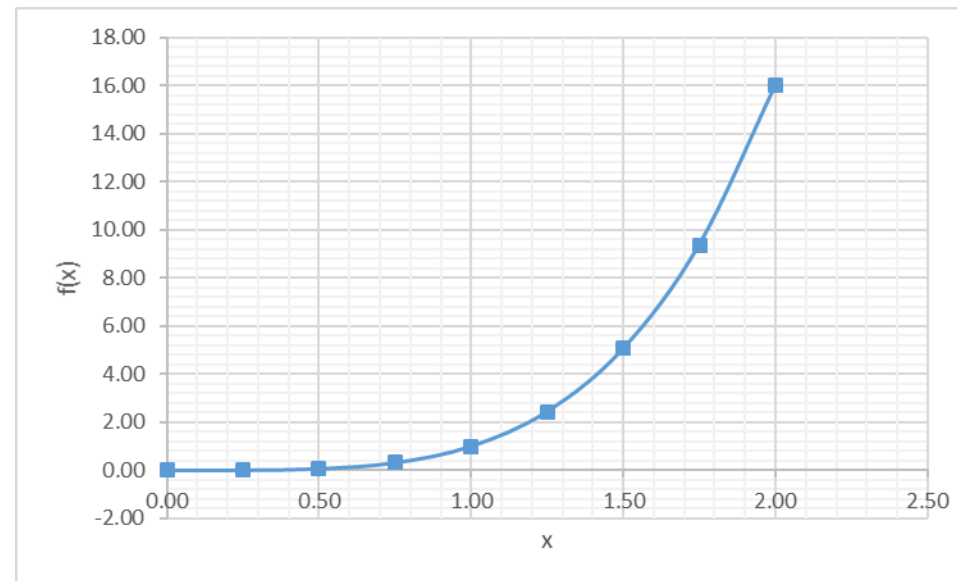
$$\begin{aligned} f(x_d = +\frac{1}{\sqrt{3}}) &= (1 + \frac{1}{\sqrt{3}})^4 = 6.19031259 \\ f(x_d = -\frac{1}{\sqrt{3}}) &= (1 - \frac{1}{\sqrt{3}})^4 = 3.190967 \times 10^{-2} \end{aligned}$$

$$\begin{aligned} I &\approx 3.190967 \times 10^{-2} + 6.19031259 \\ &= 6.22222 \end{aligned}$$

* Here we are approximating the integral of a 4th order function with two points (two-point Gauss-Quadrature)

RECALL

	n	I
Trapezoid		
	2	9.00000
	8	6.56641
	16	6.44165
	32	6.41042
	64	6.40260
Simpsons 1/3		
	2	6.66667
	8	6.40104



5.4 Numerical Integration [Gauss Quadrature]

Higher-point Gauss Formulas

General formulation:

$$I \cong c_0 f(x_0) + c_1 f(x_1) + \cdots + c_{n-1} f(x_{n-1})$$

TABLE 22.1 Weighting factors c and function arguments x used in Gauss-Legendre formulas.

Points	Weighting Factors	Function Arguments	Truncation Error
2	$c_0 = 1.0000000$ $c_1 = 1.0000000$	$x_0 = -0.577350269$ $x_1 = 0.577350269$	$\cong f^{(4)}(\xi)$
3	$c_0 = 0.5555556$ $c_1 = 0.8888889$ $c_2 = 0.5555556$	$x_0 = -0.774596669$ $x_1 = 0.0$ $x_2 = 0.774596669$	$\cong f^{(6)}(\xi)$
4	$c_0 = 0.3478548$ $c_1 = 0.6521452$ $c_2 = 0.6521452$ $c_3 = 0.3478548$	$x_0 = -0.861136312$ $x_1 = -0.339981044$ $x_2 = 0.339981044$ $x_3 = 0.861136312$	$\cong f^{(8)}(\xi)$
5	$c_0 = 0.2369269$ $c_1 = 0.4786287$ $c_2 = 0.5688889$ $c_3 = 0.4786287$ $c_4 = 0.2369269$	$x_0 = -0.906179846$ $x_1 = -0.538469310$ $x_2 = 0.0$ $x_3 = 0.538469310$ $x_4 = 0.906179846$	$\cong f^{(10)}(\xi)$



5.4 Numerical Integration [Gauss Quadrature]

Example: Solve using 3-point Gauss Quadrature

$$\int x^4 dx$$

$$= \int (1+x^2)^2 dx$$

THREE POINT GAUSS QUADRATURE

$$I \approx 5556 \cdot f(-0.7741) + 0 \cdot f(889) + f(0) + 0.5551 \cdot f(0.7746)$$

$$= 6.400085$$

ROUND OFF

I Exact = 6.4

TABLE 22.1 Weighting factors c and function arguments x used in Gauss-Legendre formulas.

Points	Weighting Factors	Function Arguments	Truncation Error
2	$c_0 = 1.0000000$	$x_0 = -0.577350269$	$\cong f^{(4)}(\xi)$
	$c_1 = 1.0000000$	$x_1 = 0.577350269$	
3	$c_0 = 0.5555556$	$x_0 = -0.774596669$	$\cong f^{(6)}(\xi)$
	$c_1 = 0.8888889$	$x_1 = 0.0$	
	$c_2 = 0.5555556$	$x_2 = 0.774596669$	

- * Provides exact values of integrals for polynomials up to degree $2n - 1$ (for n points). e.g. for 3 data points, fits polynomial of up to order 5 – should be good for 4th order

Carrying more decimal places

Gauss Quadrature			x_d	$(1+x_d)^4$	
			$f(x_d)$		
c0	0.555555555556	x0	-0.7745966690	0.002581	0.001434065
c1	0.888888888889	x1	0.0000000000	1	0.888888889
c2	0.555555555556	x2	0.7745966690	9.917419	5.509677043
					6.40000000

5.4 Numerical Integration [Gauss Quadrature]

Example: falling parachutist

$$d = \frac{gm}{c} \int_0^{10} [1 - e^{-(c/m)t}] dt$$

$$d = \frac{gm}{c} \int_0^{10} [1 - e^{-(\frac{c}{m})(5+5t)}] 5 dt$$

$$f(td) = \frac{gm}{c} [1 - e^{-\frac{c}{m}(5+5td)}] 5$$

where $g = 9.8$, $c = 12.5$, and $m = 68.1$

Two-point estimate = 290.0145

Three-point estimate = 289.4393

Four-point estimate = 289.4352

Five-point estimate = 289.4351

Six-point estimate = 289.4351

Gauss Quadrature (3 point)			f(td)		
c0	0.555555555556	x0	-0.7745966690	49.88612	27.7145107
c1	0.888888888889	x1	0.0000000000	160.3288	142.5145121
c2	0.555555555556	x2	0.7745966690	214.5785	119.2102745
g	9.8				289.43929731
m	68.1				
c	12.5				

Quadrature
(transformation)



$$x = \frac{(b+a) + (b-a)x_d}{2}$$

$$dx = \frac{b-a}{2} dx_d$$

$$x = \frac{(10+0) + (10-0)x_d}{2}$$

$$= 5 + 5x_d$$

$$dx = \left(\frac{10-0}{2} \right) dx_d$$

$$= 5 dx_d$$

Exact solution: 289.4351

5.5 Numerical Differentiation

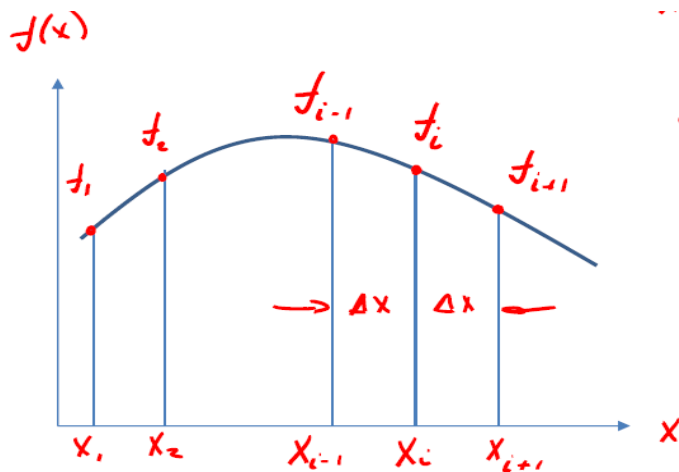
Section 23.1, 23.2 (+ some derivations in Chapter 4)

Numerically differentiation is used to approximate the derivatives of functions.

Beneficial for analyzing discrete data and functions.

For a uniform interval $\Delta x = x_{i+1} - x_i$
at an arbitrary location x_i we want to
approximate the derivative of the function

OUR GOAL IS
TO RELATE



$$\left[\frac{df}{dx} \Big|_{x_i} \quad \frac{d^2 f}{dx^2} \Big|_{x_i} \right]$$

UNKNOWN

TO VALUES OF $f(x)$
KNOWN

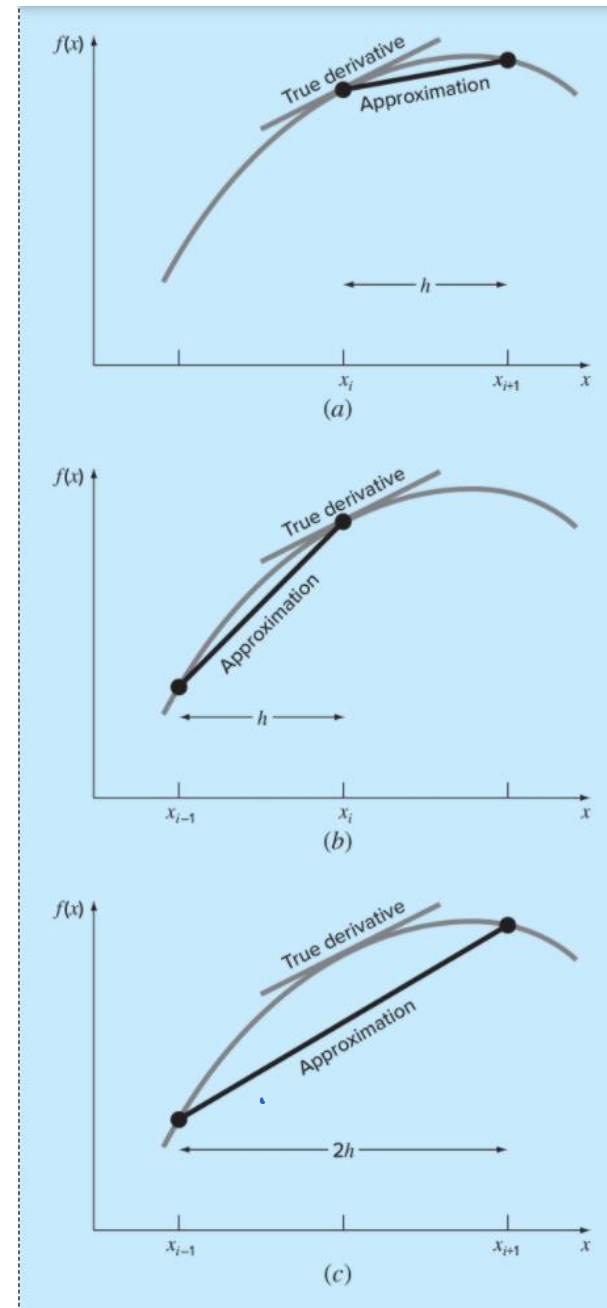


FIGURE 4.6 Graphical depiction of (a) forward, (b) backward, and (c) centered finite-divided-difference approximations of the first derivative.

Aside: Taylor Series

- Zero order approximation
- First order approximation
- Second order approximation

$$f(x_{i+1}) \cong f(x_i)$$

May be okay if the slope is small,
or points very close together

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(x_i)}{2!} (x_{i+1} - x_i)^2$$

- Taylor series
 - Any smooth function can be approximated by a polynomial
 - If a function and its first $n+1$ derivatives are continuous on an interval containing a and x , then the value of the function at x is given by:

$$\begin{aligned} f(x) = & f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 \\ & + \frac{f^{(3)}(a)}{3!}(x - a)^3 + \dots \\ & + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n \end{aligned}$$

5.5 Numerical Differentiation

Forward Difference – uses forward values of the function f_i , f_{i+1} , f_{i+2} etc to approximate the derivative at x_i .

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} + O(x_{i+1} - x_i)$$

$$f'(x_i) = \frac{\Delta f_i}{h} + O(h)$$

Δf_i is referred to as the *first forward difference* and h is called the *step size*

Consider a Taylor series expansion

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + \underbrace{\frac{h}{2!} f''(x_i) + \frac{h^2}{3!} (\dots) + \dots}_{\text{HIGHER ORDER TERMS}}$$

SINCE $h \ll 1$

WE CAN NEGLECT H.O.T.

AND THE ERROR $E \propto h = O(h)$

→ FIRST ORDER APPROXIMATION, OF ORDER h

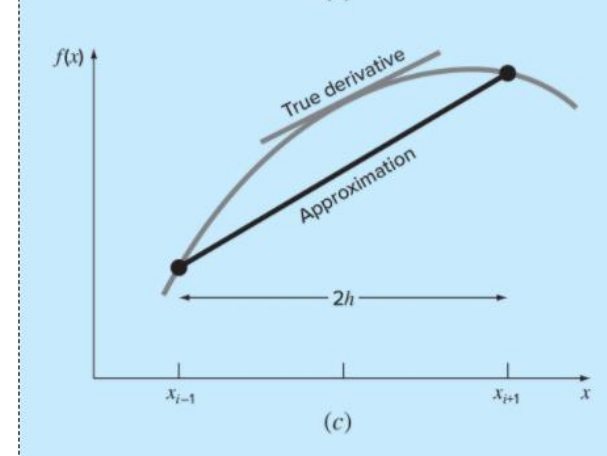
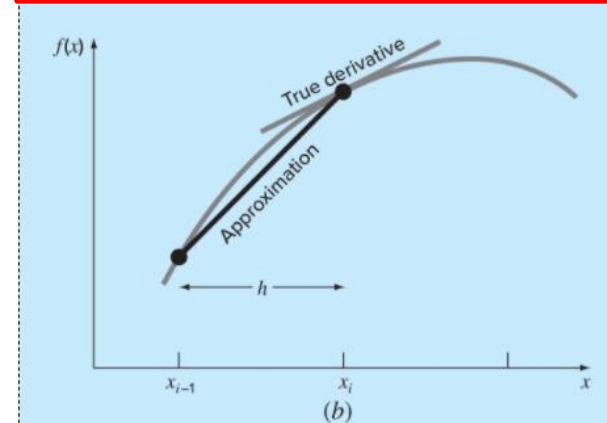
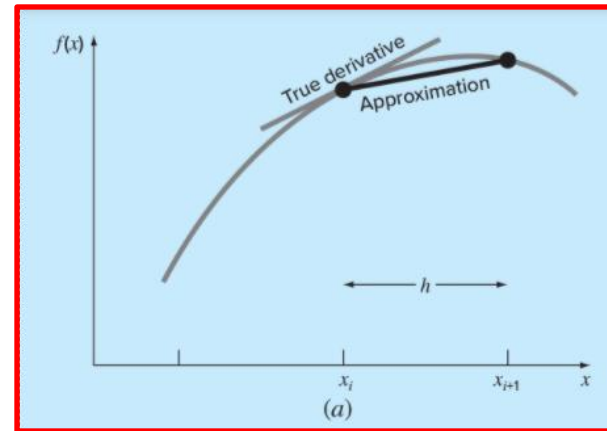


FIGURE 4.6 Graphical depiction of (a) forward, (b) backward, and (c) centered finite-divided-difference approximations of the first derivative.

5.5 Numerical Differentiation

Backward Difference – a Taylor series for the $i-1$ term:

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - \dots$$

and truncate after the first derivative:

$$f'(x_i) \cong \frac{f(x_i) - f(x_{i-1}))}{h} = \frac{\nabla f_i}{h}$$

where the error is $O(h)$, and ∇f_i is referred to as the *first backward difference*

$(x_{i-1} - x_i)$ gives a minus sign
for second and fourth terms

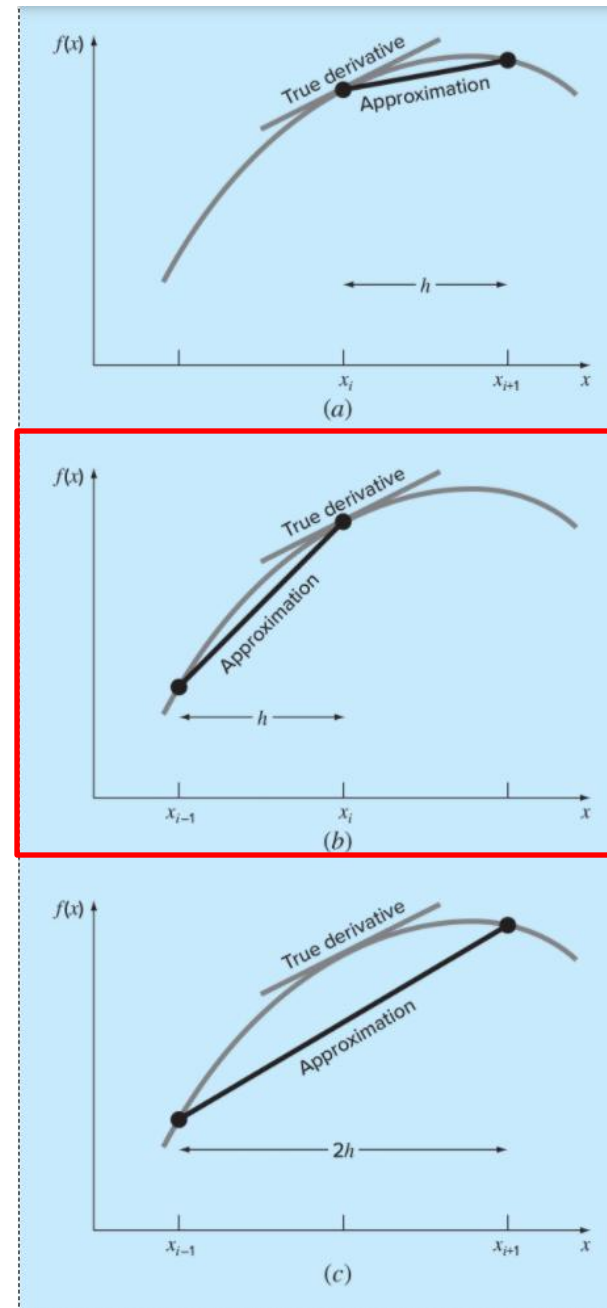


FIGURE 4.6 Graphical depiction of (a) forward, (b) backward, and (c) centered finite-difference approximations of the first derivative.

5.5 Numerical Differentiation

Centred Difference – a Taylor series for the $i+1$ and $i-1$ terms:

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots \quad [\text{Forward difference}]$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - \dots \quad [\text{Backward difference}]$$

and subtract:

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + \frac{2f^{(3)}(x_i)}{3!}h^3 + \dots$$

solve for first derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} - \frac{f^{(3)}(x_i)}{6!}h^2 - \dots$$

or

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} - O(h^2)$$

↑ SECOND ORDER METHOD

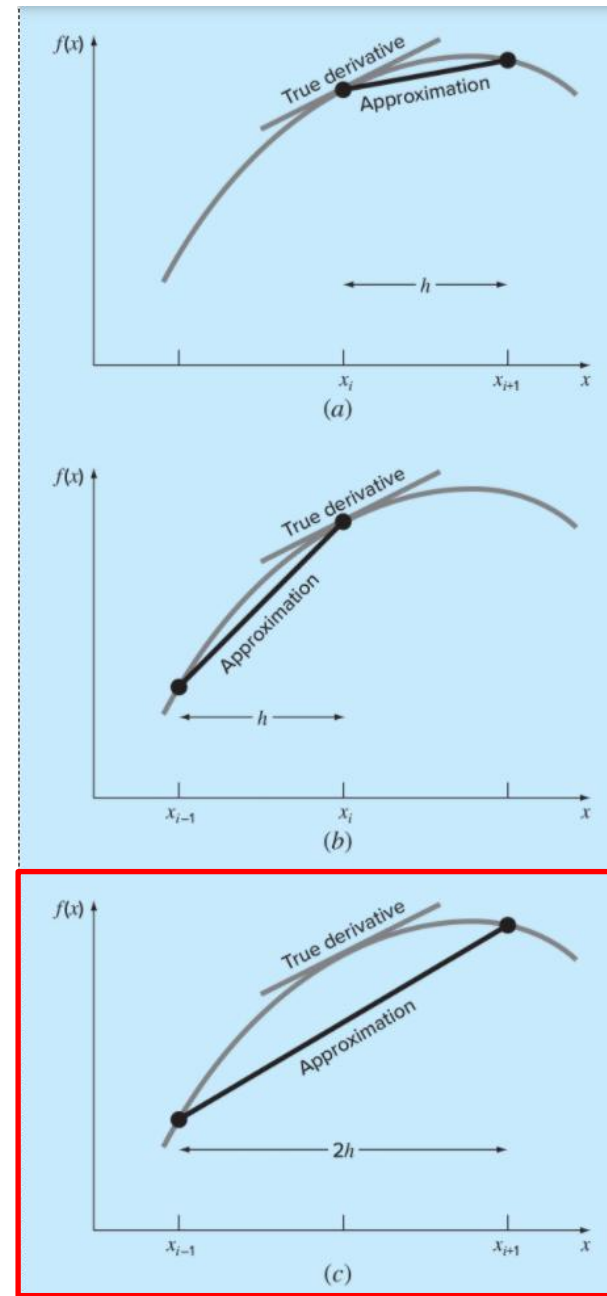


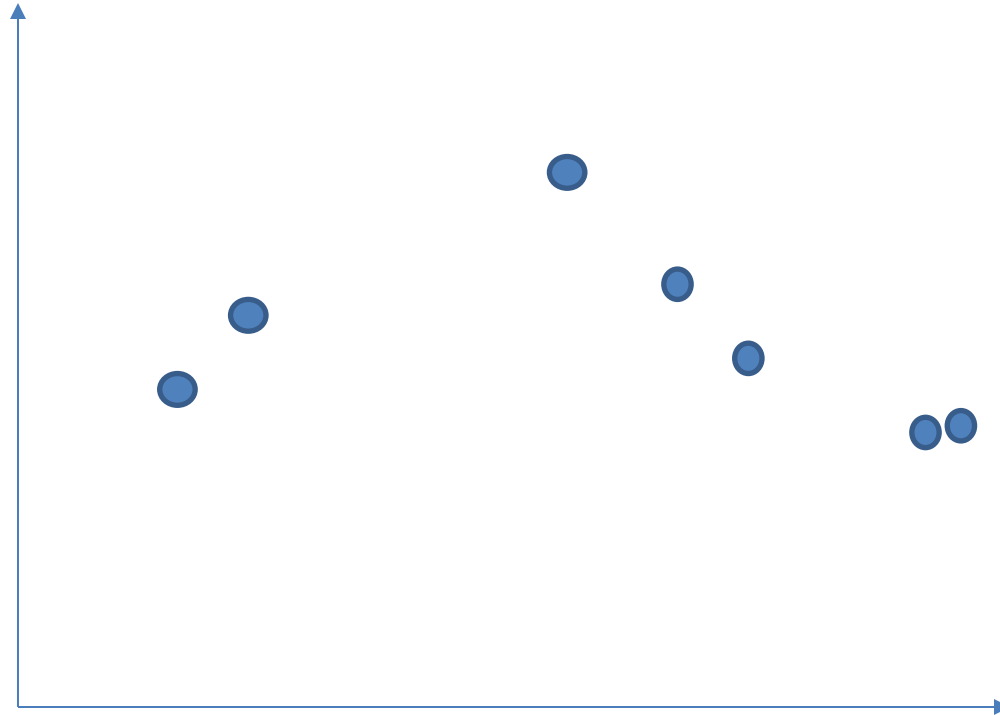
FIGURE 4.6 Graphical depiction of (a) forward, (b) backward, and (c) centered finite-difference approximations of the first derivative.

5.5 Numerical Differentiation

Unequally-spaced data

Often useful for experimental data

- Fit a polynomial to the data (piecewise), and take the derivative.
- Fit a spline and take the derivative of the spline



5.5 Numerical Differentiation [Example]

Example 5.5

Use forward and backward difference approximations of $O(h)$ and a centered difference approximation of $O(h^2)$ to estimate the first derivative of

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.25$$

at $x = 0.5$ using a step size, $h = 0.5$. Repeat the computation using $h = 0.25$.

Exact Sol'n

$$f'(0.5) = -0.9125$$

For $h = 0.5$

$$x_{i-1} = 0$$

$$f(x_{i-1}) = 1.2$$

$$x_i = 0.5$$

$$f(x_i) = 0.925$$

$$x_{i+1} = 1.0$$

$$f(x_{i+1}) = 0.2$$

$$f'(0.5)_{FD} = -1.45 \quad |E_r| = 58.9\%$$

$$f'(0.5)_{BD} = -0.55 \quad |E_r| = 39.7\%$$

$$f'(0.5)_{CD} = -1.0 \quad |E_r| = 9.6\%$$

For $h = 0.25$

$$x_{i-1} = 0.25$$

$$f(x_{i-1}) = 1.1035$$

$$x_i = 0.5$$

$$f(x_i) = 0.925$$

$$x_{i+1} = 0.75$$

$$f(x_{i+1}) = 0.6363$$

$$f'(0.5)_{FD} = -1.155 \quad |E_r| = 26.5\%$$

$$f'(0.5)_{BD} = -0.714 \quad |E_r| = 21.7\%$$

$$f'(0.5)_{CD} = -0.934 \quad |E_r| = 2.4\%$$

5.5 Numerical Differentiation [Second Derivative]

Second derivative $f''(x)$ – calculating the change in slope requires three points

$$f(x_{i+2}) = f(x_i) + f'(x_i)(2h) + \frac{f''(x_i)}{2!}(2h)^2 + \dots \quad (1) \quad [2h - \text{for two intervals to get to } x_{i+2}]$$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots \quad (2)$$

$$(1) - 2 \times (2)$$

$$f(x_{i+2}) - 2f(x_{i+1}) = -f(x_i) + f''(x_i)h^2 + \dots$$

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$$

← SECOND FORWARD DIVIDED D, FREQUENCY

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2} + O(h)$$

← BACKWARD

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} + O(h^2)$$

← CENTRAL

5.5 Numerical Differentiation [Improved Accuracy, Forward Difference]

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

Error

$$O(h)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$$

$$O(h^2)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3}$$

$$O(h)$$

$$f'''(x_i) = \frac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3}$$

$$O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4}$$

$$O(h)$$

$$f^{(4)}(x_i) = \frac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4}$$

$$O(h^2)$$

FIGURE 23.1 Forward finite-divided-difference formulas: Two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

5.5 Numerical Differentiation [Improved Accuracy, Backward Difference]

First Derivative

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1}))}{h}$$

Error

$$O(h)$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3}))}{h^2}$$

$$O(h^2)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_i) - 3f(x_{i-1}) + 3f(x_{i-2}) - f(x_{i-3}))}{h^3}$$

$$O(h)$$

$$f'''(x_i) = \frac{5f(x_i) - 18f(x_{i-1}) + 24f(x_{i-2}) - 14f(x_{i-3}) + 3f(x_{i-4}))}{2h^3}$$

$$O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_i) - 4f(x_{i-1}) + 6f(x_{i-2}) - 4f(x_{i-3}) + f(x_{i-4}))}{h^4}$$

$$O(h)$$

$$f^{(4)}(x_i) = \frac{3f(x_i) - 14f(x_{i-1}) + 26f(x_{i-2}) - 24f(x_{i-3}) + 11f(x_{i-4}) - 2f(x_{i-5}))}{h^4}$$

$$O(h^2)$$

FIGURE 23.2 Backward finite-divided- difference formulas: Two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

5.5 Numerical Differentiation [Improved Accuracy, Centred Difference]

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Error

$$O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$$

$$O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$$

$$O(h^4)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2}))}{2h^3}$$

$$O(h^2)$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3}))}{8h^3}$$

$$O(h^4)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{h^4}$$

$$O(h^2)$$

$$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) - 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) - f(x_{i-3}))}{6h^4}$$

$$O(h^4)$$

FIGURE 23.3 Centered finite-divided-difference formulas: Two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

5.5 Numerical Differentiation

Revisit Example 5.5 for $h=0.25$, investigate higher accuracy derivatives for $x=0.5$

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.25$$

$$\text{recall } f'_{\text{exact}}(0.5) = -0.9125$$

x	$f(x)$
$x_{i-2} = 0$	1.2
$x_{i-1} = 0.25$	1.104
$x_i = 0.5$	0.925
$x_{i+1} = 0.75$	0.636
$x_{i+2} = 1.0$	0.2

Forward difference $O(h^2)$

$$\begin{aligned} f'(x_i) &= \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h} \\ &= \frac{-0.2 + 4(0.636) - 3(0.925)}{2(0.25)} \\ &= -0.859 \quad (5.8\%) \end{aligned}$$

Backward difference $O(h^2)$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{2h} = \frac{3(0.925) - 4(1.104) + 1.2}{2(0.25)} = -0.878 \quad (3.8\%)$$

Centred difference $O(h^4)$

$$\begin{aligned} f'(x_i) &= \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h} \\ &= \frac{-0.2 + 8(0.636) - 8(1.104) + 1.2}{12(0.25)} \\ &= -0.9125 \quad (\approx 0\%) \end{aligned}$$